

INTRODUCTION TO DATA MANAGEMENT

PROJECT REPORT

(Project Semester January-April 2025)

Drug Report Dashboard: Accidental Drug Deaths –US (2012-2020)

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Programme and Section : CSE, KM005

Course Code : INT 217

Under the Guidance of

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CERTIFICATE

This is to certify that Dev Karan bearing Registration no. 12321910 has completed INT 217 project titled, “Drug Report Dashboard: Accidental Drug Deaths –US(2012-2020)” under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Baljinder Kaur

Signature and Name of the Supervisor

Designation of the Supervisor

School of

Lovely Professional University

Phagwara, Punjab.

Date:

DECLARATION

I, Dev Karan , student of Introduction To Data Management under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 12/04/2025

Signature

Registration No. 12321910

Dev Karan

1. Introduction

This report provides a comprehensive analysis of drug-related deaths documented in the dataset titled "**Dev's Dashboard Excel INT217 CA 2.xlsx**." The data comprises records of accidental drug-related deaths from 2012 to 2023, as reported by medical examiners and health authorities.

The dataset includes rich detail about each case, such as demographic information (age, sex, race), substances involved (e.g., Fentanyl, Heroin, Cocaine), location data (city, county, residence), and temporal patterns (date and year of death)

The primary focus of this report is to identify **trends and patterns in drug involvement**, explore the **demographic impact** of various substances, and assess the **geographic distribution** of drug-related fatalities. By analyzing attributes like:

- **Substance involvement** (Fentanyl, Heroin, Cocaine, etc.)
- **Victim demographics** (Age, Sex, Race)
- **Date and year of death**
- **Place of occurrence and residence**

...this dashboard aims to uncover key insights into overdose patterns, multi-drug interactions, and high-risk groups.

In addition, the report highlights patterns in **age distribution**, **drug combinations**, and **racial disparities** in substance-related mortality, offering a foundation for **data-driven awareness** and **preventive policy design**.

2. Source of Dataset

The data I used in this project came from a trusted website called data gov, an usa website (<https://data.gov>). It has a lot of information about accidental death drug cases across united states.

You can check out the same dataset I used by going to this link:

<https://catalog.data.gov/dataset/accidental-drug-related-deaths-2012-2018>

3. Dataset Preprocessing

- **Rows and Columns:** The dataset contains **13,419 records** and **29 columns**, covering detailed information on accidental drug-related deaths.
- **Missing Data:** A few fields such as **City**, **Residence City**, and **Residence County** have missing or blank values, which may reflect unavailable information in official reports.
- **Data Types:** The dataset includes a mix of **numerical fields** (like Age), **dates** (Date of Death), and **categorical values** (Sex, Race, Drug Names, etc.).
- **Date Formatting:** Dates are consistently formatted in the **'Date' column**, allowing for easy filtering and time-based analysis.

4. Analysis on Dataset (for each objective)

Chart 1: Gender Distribution by Age

i. General Description

This chart visualizes the **distribution of drug-related deaths by gender across different age groups**. It helps understand how male and female populations are impacted at various stages of life.

ii. Specific Requirements

- Use **Sex** as the primary category (Male/Female)
- Use **Age** as a grouping or axis variable
- Count the **number of cases** for each gender across age ranges

iii. Analysis Results

- **Males** account for a higher number of deaths across most age groups.
- The **age range between 25–44 years** shows a peak in fatalities, especially among males.
- Female-related deaths are significantly lower but follow a similar pattern in age distribution.

iv. Visualization

- **Pie Chart**
- X-axis: Age
- Series: Sex (Male, Female)
- Y-axis: Count of Cases
- Visually separates gender comparison and highlights vulnerable age segments

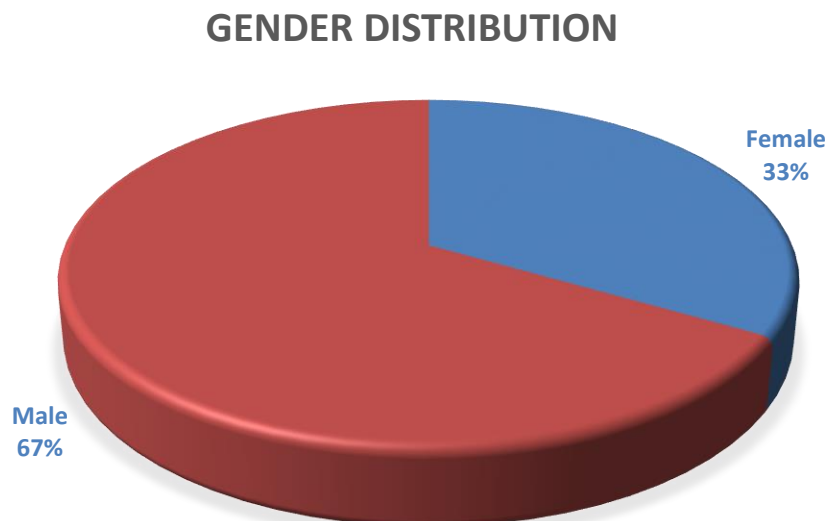


Chart 2: Death Incidents by Race

i. General Description

This chart presents a view of **drug-related death incidents categorized by race**, using the **count of age** as a proxy for the number of cases per racial group.

ii. Specific Requirements

- Use **Race** as the primary grouping axis
- Use **Age** to count the number of cases (since each age represents a unique individual case)
- Filter out blank or unknown race entries if necessary for clarity

iii. Analysis Results

- **White individuals** represent the largest group affected by drug-related deaths.
- **Black/African American** individuals follow in case count, while other racial groups show fewer reported incidents.
- This chart highlights racial disparities in drug overdose mortality, which can reflect broader social and healthcare factors.

iv. Visualization

- **3D Clustered Bar Chart**
- X-axis: Race
- Y-axis: Count of Age (used as case count)
- Offers a clear comparative view of how drug-related deaths are distributed across racial groups

Death Incidents by Race

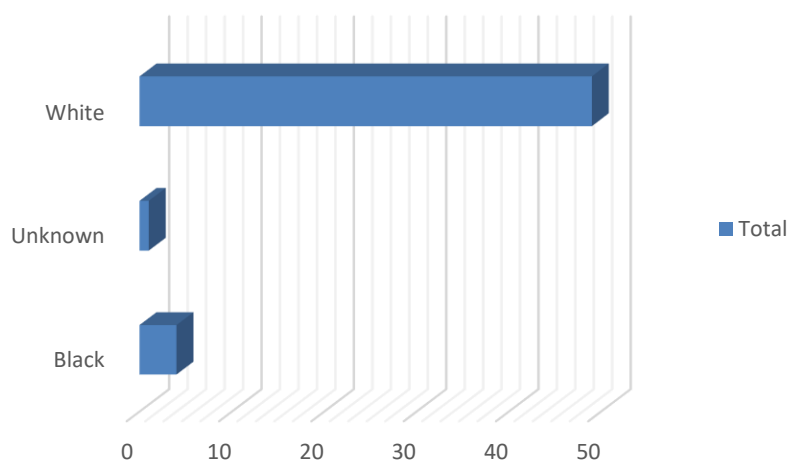


Chart 3: Most Common Races by Location

i. General Description

This chart analyzes **which racial groups are most represented in drug-related deaths across different cities** (based on the residence of the deceased). It gives insight into how racial demographics and geographic areas intersect in the context of drug-related fatalities.

ii. Specific Requirements

- Group data by **Residence City**
- Use **Race** as a series/category dimension
- Count the number of records for each race in each city (use **Race** column for the count)
- Handle blank residence cities or unknown races as needed

iii. Analysis Results

- Some cities have a dominant racial demographic in reported deaths (e.g., **White individuals in certain urban areas**).
- Racial diversity in drug-related deaths varies significantly by location—some cities show more balance among racial groups, while others show concentration.
- This chart reveals **localized racial impact**, which can inform targeted community interventions.

iv. Visualization

- **Stacked Column Chart**
- X-axis: Residence City
- Series: Race
- Y-axis: Count of Race (used as the measure of cases)
- Shows side-by-side racial breakdowns per city, making it easy to compare across regions

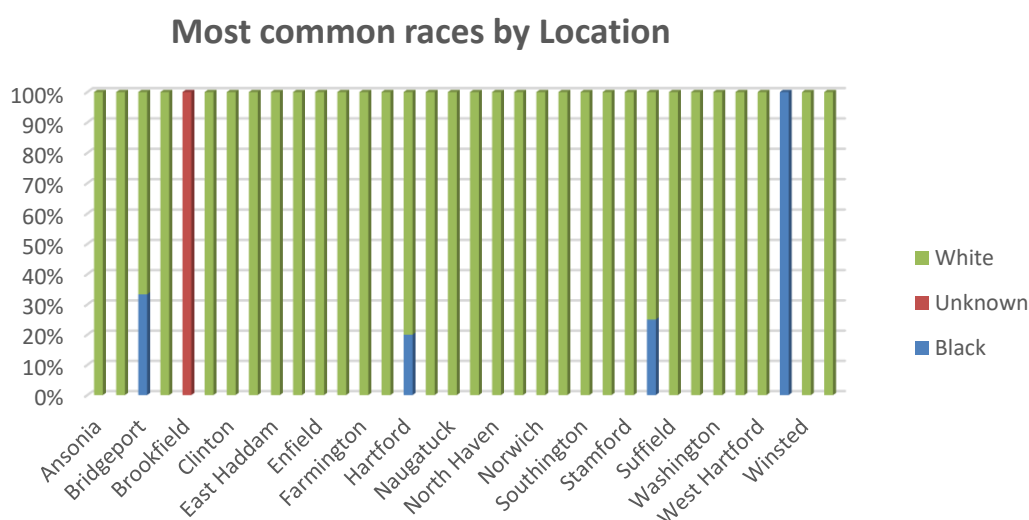


Chart 4: Drug Use by Location and Year

i. General Description

This chart tracks **how drug-related deaths vary by U.S. state and year**, helping to identify patterns in drug use across regions over time. It gives a temporal and geographical perspective on the drug crisis.

ii. Specific Requirements

- Use **Residence State** as the primary axis
- Use **Year** (extracted from the Date column) as the series/category
- Use **Count of Age** as a proxy for number of deaths (each row represents one person, so counting age gives total incidents)
- Clean or group blank/unknown residence states if needed

iii. Analysis Results

- Certain states (like **Connecticut**, where most data points originate) show **consistently high numbers across years**
- The data reveals whether deaths are **increasing, decreasing, or stable** over time for each state
- Highlights **regional trends**—some states may see spikes in certain years, possibly due to changes in drug availability or law enforcement patterns

iv. Visualization

- **Clustered Column Chart**
- X-axis: Residence State
- Series: Year
- Y-axis: Count of Age (proxy for case count)
- Useful for spotting year-over-year trends and geographic hot spots

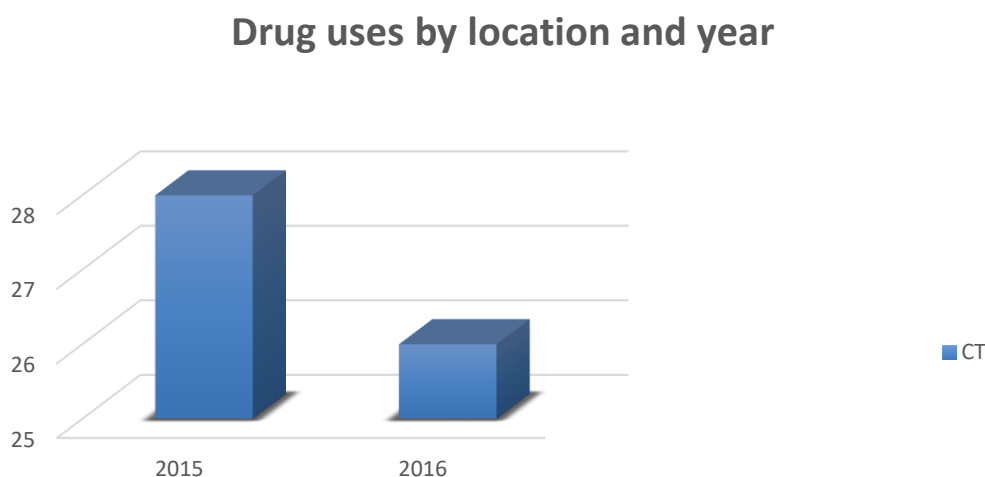


Chart 5: Percentage of Male vs Female Involvement in Races

i. General Description

This chart illustrates the **gender distribution of drug-related deaths within each racial group**, showing the percentage of **male vs. female** cases. It helps to reveal how gender dynamics differ among various races in the dataset.

ii. Specific Requirements

- Group data by **Race**
- Use **Sex (Male/Female)** as the category for comparison
- Count the number of records per gender within each race (using **Count of Sex**)
- Display values as **percentages of the total per race** (showing proportions, not raw counts)
- Optionally filter out “Unknown” sex or race if present

iii. Analysis Results

- Across most races, **males make up the majority of drug-related deaths**
- Some races have a relatively **higher female percentage**, but the gender gap is consistent
- This view helps identify if certain racial groups have a **disproportionate male or female representation**

iv. Visualization

- **3D Line Chart**
- X-axis: Race
- Series: Sex (Male, Female)
- Y-axis: Percentage (0–100%)
- Visually emphasizes the gender proportions within each race group

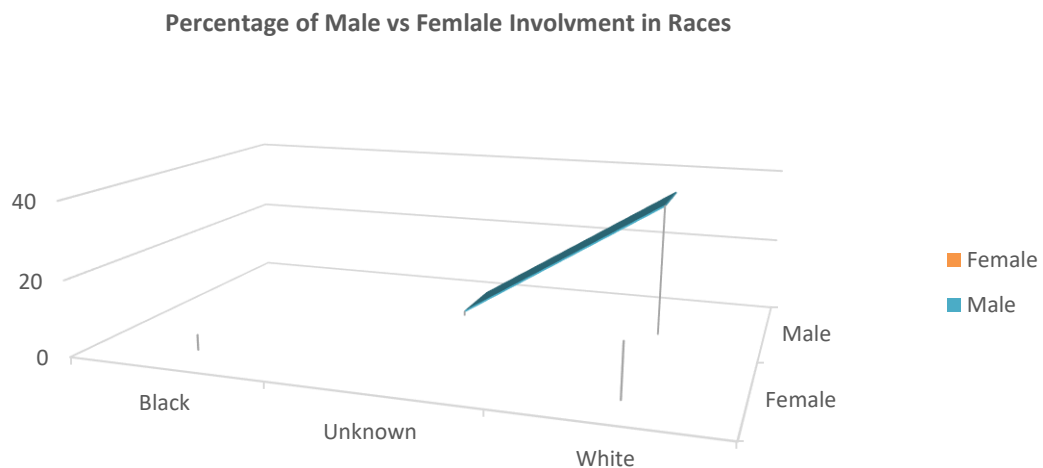


Chart 6: Average of Age by Drug Involvement

i. General Description

This chart analyzes the **average age of individuals who were involved with different drugs**, specifically **Heroin, Cocaine, and Fentanyl**. It helps identify which substances are more commonly linked to younger or older age groups in drug-related deaths.

ii. Specific Requirements

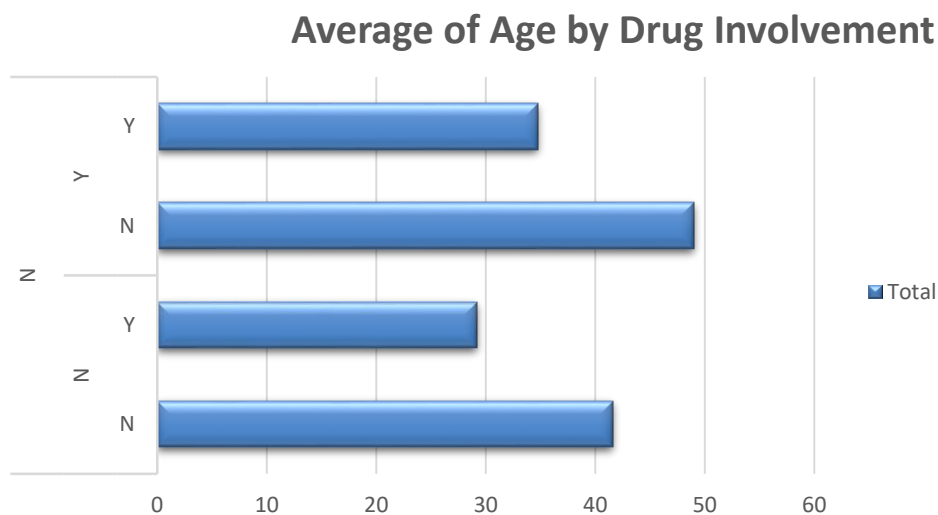
- Group by **drug type** (Heroin, Cocaine, Fentanyl)
- Use **Average of Age** as the value
- Filter only rows where each drug was marked as involved (Y or similar flag)
- Display results as **separate bars** for easy comparison

iii. Analysis Results

- The **average age varies across substances**:
 - **Fentanyl** tends to involve slightly **younger individuals**
 - **Cocaine** shows a **higher average age** than Heroin
 - This insight suggests demographic differences in substance use or availability
- Could reflect generational trends in drug preference or access

iv. Visualization

- **Combine Chart**
- X-axis: Drug Type (Heroin, Cocaine, Fentanyl)
- Y-axis: Average of Age
- Clean and minimal layout emphasizes comparison of average ages by drug type



5. Conclusion

The drug overdose analysis dashboard provides a comprehensive understanding of the key factors contributing to accidental drug-related deaths. By examining patterns across age, gender, race, location, drug type, and year, the analysis uncovers several critical insights.

Drug-related deaths show notable variation across demographic groups, with males and White individuals consistently representing the highest proportion of cases. Certain residence cities and states report significantly higher numbers of incidents, indicating geographic hotspots that may benefit from targeted public health interventions and resource allocation.

The average age of individuals involved varies depending on the type of drug, with substances like Fentanyl often affecting younger individuals, while others like Cocaine are more prevalent among

older age groups. These trends reflect shifts in drug availability and usage patterns across age cohorts.

While deaths occur throughout the year, some years and locations exhibit noticeable spikes, suggesting emerging crises or the impact of policy and enforcement changes. The gender and race breakdowns further highlight disparities in the impact of substance abuse, emphasizing the need for inclusive, community-specific harm reduction strategies.

This dashboard serves as a vital tool for health officials, policymakers, and support organizations to identify high-risk populations, track trends over time, and prioritize efforts to reduce fatal outcomes from drug use.

6. Future Scope

The current drug overdose analysis dashboard delivers valuable insights into patterns and contributing factors behind accidental drug-related deaths. However, there is considerable potential to further enhance its capabilities and societal impact through future developments.

One major area for advancement is the integration of real-time public health and emergency response data, which could enable early detection of overdose spikes and support timely intervention. Real-time feeds from hospitals, EMS reports, or prescription monitoring programs would provide a dynamic layer of situational awareness, allowing health officials to respond more proactively to emerging drug crises.

Incorporating machine learning models could further elevate the dashboard's effectiveness by predicting high-risk demographic groups, geographic regions, and time periods. By learning from historical patterns in drug involvement, age, race, and location, these models could identify subtle trends and recommend targeted preventative strategies.

Additionally, expanding the dataset to include variables such as mental health history, socioeconomic background, housing status, prescription drug records, and co-occurring substance use would provide a more comprehensive and nuanced view of overdose causes. This would support more informed public health initiatives and community outreach efforts.

By embracing these enhancements, the dashboard could evolve into a powerful decision-support tool that not only monitors trends but also predicts and helps prevent future drug-related deaths.

7. References

- “Accidental Drug Deaths” [Online]. Available: <https://catalog.data.gov/dataset/accidental-drug-related-deaths-2012-2018>
- Tools used: Microsoft Excel .

❖ LinkedIn

<https://www.linkedin.com/feed/update/urn:li:activity:7317228012389781504/>

The screenshot shows a LinkedIn post from a user named 'Portfolio Project'. The post features a dashboard titled 'Accidental Drug Deaths-US(2012-2020)' with several charts and tables. The dashboard includes a pie chart for 'Accidental Drug Deaths by State', a bar chart for 'Accidental Drug Deaths by Age Group', a line chart for 'Accidental Drug Deaths by Year', a bar chart for 'Accidental Drug Deaths by Drug Type', a bar chart for 'Accidental Drug Deaths by Gender', and a table for 'Accidental Drug Deaths by State'. The post has 18 comments and 34 reactions. The user's profile picture is a blue square with a white 'in' logo. The post is shared on a mobile device, as indicated by the status bar at the top showing the time as 10:10 AM and the battery level at 100%.